Project Status & Information

Submersible Pump Power Abnormality Monitoring & Protection System

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1. Project Goal

Design and implement a real-time electronic monitoring and protection system for TYPE 4 series three-phase submersible drainage pumps (10–30 A). The system continuously measures electrical parameters, detects dangerous power abnormalities, and automatically disconnects the pump via the existing contactor to prevent motor burnout. The complete work will be published as a professional open-source repository for an embedded systems Master's application portfolio.

2. Background & Context

The developer works as a Software Specialist in the IT department of an industrial pump manufacturer (previously completed a 6-month internship at the same company). The project bridges industrial experience with embedded systems skills. The pumps are 3-phase industrial drainage pumps (TYPE 4 series) with power ratings typically in the 8–18 HP range. An existing electromechanical DOL starter panel is already available and will be reused.

3. Technical Requirements

- Supply: 3-phase, current range 10–30 A
- Primary protections (priority order):

1. Over-voltage / Under-voltage
2. Over-current
3. Phase loss / severe phase imbalance

- Action on dangerous condition: Open contactor coil circuit → pump stops
- Safety rule: Manual reset required after every trip (no automatic restart)
- Local indication: Status LEDs + OLED display showing state and trip reason
- Low-budget approach while remaining professional and documentable

4. Existing Hardware (Already Available)

Item	Details
Contactor	Hyundai HGC 25 – Ith = 45 A, AC-3, Coil = 220–240 V AC
Manual Motor Starter	Hyundai HMMS-32K – adjustable 24–32 A thermal + magnetic protection
Control Station	START / STOP push buttons + Emergency Stop + status indicators
Other	DIN-rail panel, wiring, thermal overload already integrated

5. Electronic Parts Ordered

Qty	Component	Specification
1	ESP32 Dev Board	ESP32-WROOM-32 (USB)
3	Voltage Sensor	ZMPT101B (isolated)
3	Current Transformer	YHDC SCT-013-030 (30 A / 1 V) or SCT-013-050 (50 A / 1 V)

1	Relay Module	5 V opto-isolated, 10 A / 250 V AC
1	OLED Display	0.96″ 128×64 SSD1306 I2C
1	ADS1115 ADC	16-bit I2C (higher accuracy)
1 set	LEDs + Resistors	Red / Yellow / Green + 220 Ω
1	Reset Button	Momentary Normally-Open
1–2	Breadboard	400-point prototyping board

6. Progress Completed So Far

- Project goal and scope defined
- Existing industrial panel fully identified (HGC 25 contactor + HMMS-32K + control station)
- Contactor coil voltage confirmed: 220–240 V AC
- Hardware architecture decided (ESP32 + ZMPT101B + SCT-013 + relay + OLED + ADS1115)
- All essential electronic components ordered
- Safety concept established (opto-isolated relay controls contactor coil, manual reset required)
- Project stages roadmap and shopping list documents created

7. Next Steps (While Waiting for Parts)

1. Create the public GitHub repository and folder structure
2. Write README.md and requirements.md
3. Draw the connection diagram (schematic)
4. Prepare PlatformIO / Arduino environment for ESP32
5. After parts arrive: assemble, calibrate, test on real pump under supervision

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This document records the current state of the project as of mid-September 2026. It will be updated as hardware arrives and implementation progresses.